

a) According to the authors, is it generally more promising to generate multiple 2D plots with different dimension reduction techniques to obtain a more comprehensive impression of a high-dimensional dataset, or is it better to retain a larger number of dimensions, and visualize them with a SPLOM? Which figure shows the data that is relevant to this question? Does it suggest any exceptions from the general rule?

We believe that using 2D scatterplots is the best starting point because the authors state that 2D is often "good enough" for cluster verification. They explain that 2D has the lowest time cost because a user can see everything in one static view without needing to interact or look at many different areas. If 2D doesn't work, the authors suggest that it is usually more promising to try a different dimension reduction (DR) technique in 2D before trying a SPLOM. Figure 4 illustrates this because it shows the few cases where SPLOMs actually performed better than 2D. We can see a specific exception in Figure 4(b) with the "industryIndices" dataset. In this case, the standard 2D view (1x2) failed to show the clusters, but a different view in the SPLOM (the 2x3 view) successfully revealed the classes as separate groups.

b) The paper includes the quite drastic statement that "interactive 3D rarely helps but often hurts". State at least two problems of interactive 3D visualizations in this context. Explain why, in another application context, interactive 3D visualizations (not necessarily 3D scatterplots) might still be effective. Give an example of such a case.

We see that the authors are very critical of 3D because it creates many visual problems. Two major issues they mention are occlusion, where some data points hide behind others, and depth ambiguity, which makes it hard to judge distances correctly. They also state that navigating and rotating in 3D takes too much time and can be difficult for the user. However, the authors explain that 3D can still be effective for "spatial data". We understand this means data that already has a physical 3D shape in the real world. A good example is medical imaging (like MRI scans) or architecture. In these cases, 3D is useful because we are looking at a real physical object rather than abstract numbers.

c) Why did the authors decide to perform a time-consuming manual coding of class separation rather than use an objective measure such as the distance consistency or distribution consistency that we learned about in the lecture? How did they verify the reliability of their subjective impression of class separation? Based on their results, is it possible to state which combination of dimension reduction and visual encoding worked best for each of the datasets?

We believe the authors chose manual coding because they found that automatic computer measures were not reliable. The paper states that automatic measures produced incorrect results for about half of the 816 scatterplots. To prove that their human judgments were reliable, they used a statistical test called Krippendorff's alpha and achieved a high score of 0.858. This score shows that the two human coders were very consistent and saw the same things. Finally, the authors conclude that it is not possible to pick one single best combination for every dataset. They explain that there is "no one-and-only DR solution" because different methods work better depending on the unique characteristics of each dataset.

d) SPLOMs are flexible with respect to the dimension, as long as it does not become too large. How has a suitable number of dimensions been selected in this work? How large were the resulting SPLOMs on average?

For the SPLOMs, we see that the researchers selected the number of dimensions by checking when adding more views stopped providing benefits for seeing the clusters. They set a maximum limit of 15 dimensions to keep the study manageable. On average, the resulting SPLOMs were quite small, with a

mean of 4.1 dimensions and a maximum of 7. Interestingly, even though the matrices were flexible, the study found that in 88% of cases, the coders only needed to look at 1 to 4 specific views within the matrix to understand the class structure.